

面向视觉语言模型工具集成推理的智能体强化学习扩展

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Abstract

尽管近期视觉语言模型 (VLM) 在图像理解方面表现强劲, 但它们“借助图像思考”的能力, 也就是通过多步视觉交互完成推理的能力, 仍然有限。我们提出 *VISTA-Gym*, 一个可扩展训练环境, 用于激励 VLM 获得工具集成的视觉推理能力。 *VISTA-Gym* 以统一接口整合了多种真实世界多模态推理任务 (总计来自 13 个数据集的 7 类任务), 同时提供标准化视觉工具接口 (如 *grounding*、*parsing*)、可执行交互循环、可验证反馈信号以及高效轨迹日志记录, 从而支持大规模视觉智能体强化学习。虽然近期 VLM 已展现出较强的纯文本推理能力, 但无论闭源还是开源模型, 在工具选择、调用与协同上仍然表现不足。基于 *VISTA-Gym*, 我们训练了 *VISTA-R1*, 使其能够通过多轮轨迹采样与端到端强化学习, 在智能体推理过程中交替执行工具使用与推理。我们在 11 个公开、推理密集型 VQA 基准上的大量实验表明, *VISTA-R1-8B* 相比相近规模的最强基线提升了 9.51% 到 18.72%, 说明 *VISTA-Gym* 是解锁 VLM 工具集成推理能力的有效训练场。 *VISTA-Gym* 与 *VISTA-R1* 的代码和数据见 <https://github.com/Lucanyc/VISTA-Gym>。

1. 引言

近来 VLM 在视觉问答、多模态推理和具身数学问题求解等任务上取得了显著进展 [2, 11, 16, 58]。其

中不少进展来自文本推理范式的引入, 尤其是思维链 (CoT) [63] 与强化学习 (RL) [14]: 前者把复杂任务拆解为中间文本推理步骤, 后者则通过基于结果的奖励来优化推理质量 [18, 27, 30, 32, 45, 52, 53, 77]。

但大多数现有 VLM 的推理过程仍依赖静态视觉嵌入和较浅的跨模态对齐。因此, 它们的纯文本推理往往难以捕捉真实场景中的细粒度视觉结构、空间关系和数量依赖 [13, 78]。这些局限凸显了“借助图像思考”范式 [57] 的必要性, 即让推理与视觉感知紧密耦合, 从而实现更丰富的跨模态交互和逐步视觉推理。

为了增强以视觉为中心的推理能力, 并促使 VLM 形成这种 *thinking-with-image* 行为, 近期研究提出了工具集成推理 (TIR) [15, 69, 78]。TIR 为模型提供外部工具, 例如 *grounding*、*zoom-in* 和 *search*, 帮助其进行细粒度感知和围绕对象交互的推理。尽管这一方向很有前景, 当前开源 VLM 在借助这些工具进行有效推理方面仍然明显不足, 如图 1 所示。这表明, 我们亟需一个有效的训练环境与训练策略, 让模型学会动态地选择、调用和协同视觉工具。虽然已有一些工作专门提升 VLM 的工具使用能力 [13, 56, 64, 78], 但它们往往范围较窄, 通常只覆盖特定任务或单一工具类型。与此同时, 游戏和机器人领域近期也提出了一些统一环境来训练 VLM 智能体 [8, 60]。这些平台具备可控动态、接地反馈和结构化任务空间, 在机器人学习中优势明显, 也在概念上与 TIR 有共通之处。然而, 针对开放域视觉推理的工具集成式思考训练, 尤其是面向 VLM 的统一环境, 仍然缺乏系统研究。

基于这些挑战, 我们提出 *VISTA-Gym* (Visual-

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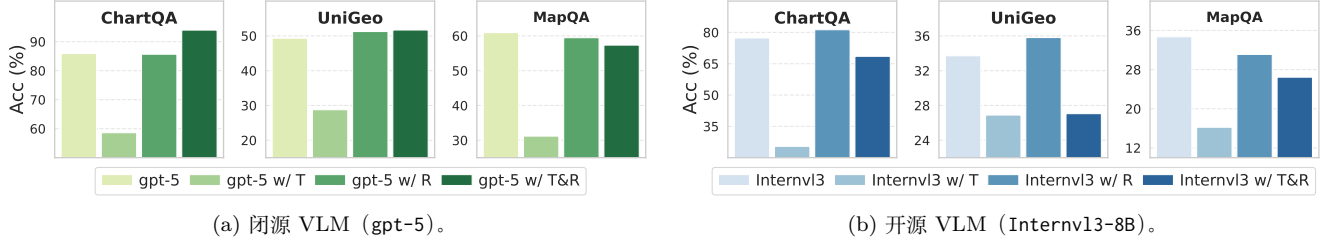


图 1. 直接给 VLM 增加工具会显著降低准确率 (w/ T), 而仅靠内在推理在复杂 VQA 上提升有限 (w/ R)。提供工具选择先验知识, 并让推理与工具执行交错进行, 可以改善性能 ($w/ T \& R$); 但这种收益对商业 VLM 具有明显任务依赖性, 而小型开源 VLM 尤其吃力。

centric Tool-integrated Agentic training environment), 这是一个可扩展的智能体训练环境, 旨在系统提升 VLM 智能体在复杂真实场景中的推理与决策能力。具体来说, VISTA-Gym 对视觉工具操作做了统一封装, 并在接收到 VLM 发出的工具调用命令后返回文本反馈。它具有以下特点:

- **多样化多模态推理任务。** VISTA-Gym 提供了覆盖 13 个公开数据集、7 类推理任务的综合 VQA 套件, 可用于训练和评估具备较强泛化与工具集成能力的智能体。
- **统一且可扩展的工具接口。** VISTA-Gym 通过标准 API 暴露 26 个预定义工具, 覆盖视觉感知、符号化操作、图表和文档理解等过程, 使高层推理能够建立在结构化中间结果之上。
- **可扩展交互基础设施。** VISTA-Gym 借助多线程、并行执行和顺序采样来加速智能体训练, 从而支持高效轨迹收集以及面向不同智能体框架的大规模自动评测。

基于 VISTA-Gym, 我们进一步提出 VISTA-R1, 一个面向稳健、工具增强推理训练的 VLM 智能体。在 5 个分布内和 6 个分布外的推理密集型 VQA 基准上, VISTA-R1-8B 相比相近规模的最强开源基线提升了 9.51% 到 18.72%。这些结果表明, VISTA-Gym 是一个有效且可扩展的方案, 可用于开发能够在复杂多步视觉问题中稳健交错使用推理与工具的 VLM 智能体。

2. 相关工作

面向 VLM 推理的强化学习。 提升 VLM 推理能力一直是活跃研究方向。[68] 通过从教师模型蒸馏推理知识

来合成推理链。受 R1 风格训练 [14] 成功启发, 一系列工作尝试借助 RL [18, 26, 29, 32, 51] 来提升 VLM 在视觉任务上的推理能力, 覆盖一般视觉理解与数学推理等场景。

VLM 中的工具集成推理强化学习。 为了更好地刻画视觉信息, 近期研究探索了“借助图像思考”的范式, 它超越了标准推理步骤, 把额外工具纳入推理过程 [17, 24]。DeepMMSearch-R1 [42] 和 MMSearch-R1 [64] 利用图像搜索工具引入外部知识; 其他工作 [13, 56, 77, 78] 则通过 *zoom-in* 等操作聚焦细粒度区域, 从而增强 grounding 与多步感知推理。ReLook [25] 还把一个 VLM 作为辅助工具, 通过跨模型交互实现智能体训练。尽管如此, 大多数现有方法仍依赖单一、专门化工具, 并局限在较窄任务域中, 因此难以泛化到更广泛的多模态推理场景。

面向智能体推理的 RL 训练环境。 为了支持智能体模型训练, 已有多种框架被提出, 包括 GEM [31]、Agent-GymRL [66]、SkyRLGym [4]、Collaborative GYM [49] 和 RAGEN [62] 等通用智能体训练框架。除此之外, 还有许多工作面向特定领域构建环境, 例如文本推理 [55]、软件工程 [12, 19, 44, 72]、机器学习工程 [43, 46, 47]、搜索与网页浏览 [9, 67, 70], 以及科学推理 [61, 71]。不过, 这些环境绝大多数都运行在纯文本设置中, 对多模态 grounding 或视觉推理支持有限。对 VLM 而言, 据我们所知, 只有 VLM-Gym [8] 和 VAGEN [60] 两个环境专门面向 VLM 训练, 分别聚焦组合式视觉游戏和具身任务。相比之下, VISTA-Gym 通过提供一个可扩展的工具集成 RL 环境, 填补了这一空白, 使得针对 VLM 的系统化视觉智能体推理训练成为可能。

表 1. Error pattern identification and distribution from 500 error samples. Note that one case may contain multiple error types.

ID	Error Type	GPT-5	InternVL3-8B
E1	Invocation schema violation (wrong function call structure)	12.8%	3.8%
E2	Invalid argument name (wrong augment name)	0.0%	44.8%
E3	Invalid argument value (wrong augment value format)	18.2%	18.0%
E4	Incorrect argument value (wrong argument value content)	39.6%	57.4%
E5	Invalid output from tool execution (wrong answer format)	25.6%	0.0%
E6	Incorrect reasoning from tool execution	28.1%	64.8%

3. 预备知识

问题设定。我们研究的是一种视觉接地问答任务，它要求模型同时对图像与文本做复杂推理，并最终输出预测 $\hat{y}$ 。模型配备一个预定义、有限的工具集合（即动作空间） $\mathcal{A}$ 。形式化地说，目标是合成一条交错包含推理步骤 g_t 与外部工具调用 $a_t \in \mathcal{A}$ 的轨迹 U 。整条解题路径可表示为 $U = (g_0, a_0, \dots, g_T, \hat{y})$ ，其中 T 表示到达最终答案所需的工具交互轮数。

VLM 中的工具集成推理。我们把 VLM 中的 TIR 形式化为一个部分可观测马尔可夫决策过程 (POMDP)，记为 $\langle \mathcal{S}, \mathcal{O}, \mathcal{A}, \mathcal{I}, \mathcal{P}, \mathcal{R} \rangle$ ，其中 $\mathcal{I}$ 、 $\mathcal{S}$ 、 $\mathcal{O}$ 、 $\mathcal{A}$ 和 $\mathcal{R}$ 分别表示指令、环境状态、智能体观测、动作和奖励空间， $\mathcal{P} : \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{S}$ 表示确定性的状态转移函数。遵循 ReAct [74]，我们约束智能体的输出先生成推理思考，再生成动作。给定问题描述 $x \in \mathcal{I}$ 、历史以及当前反馈，在第 t 轮，智能体先生成思考 $g_{t+1} \sim \pi_\theta(\cdot|x, g_1, a_1, o_1, \dots, g_t, a_t, o_t)$ ，随后生成动作 $a_{t+1} \sim \pi_\theta(\cdot|x, g_1, a_1, o_1, \dots, g_t, a_t, o_t, g_{t+1})$ 。在执行 a_{t+1} 之后，环境经由 $\mathcal{P}$ 转移到新状态 s_{t+1} ，并给出新的部分观测 $o_{t+1} \sim \mathcal{O}(\cdot|s_{t+1})$ 。因此整条轨迹可写为

$$\tau = (g_0, a_0, o_0, \dots, o_{T-1}, g_T, \hat{y}) \sim \pi_\theta(\tau|x),$$

其中 $\hat{y}$ 是由模型最终生成得到的答案， π_θ 可以分解为

$$\begin{aligned} \pi_\theta(\tau|x) &= \pi_\theta(g_T|x, c_{T-1}) \cdot \prod_{t=0}^{T-1} \pi_\theta(g_t, a_t|x, c_{t-1}) \\ &= \pi_\theta(g_T|x, c_{T-1}) \cdot \prod_{t=0}^{T-1} \pi_\theta(a_t|x, c_{t-1}, g_t) \cdot \pi_\theta(g_t|x, c_{t-1}), \end{aligned}$$

其中 $c_{t-1} = (g_0, a_0, o_0, \dots, g_{t-1}, a_{t-1}, o_{t-1})$ 表示截至第 $t-1$ 步的交互历史。

探索性实验。我们在三个 VQA 任务 [5, 7, 38] 上，使用闭源与开源 VLM 做了探索性实验。如图 1 所示，如

果直接给基础模型开放工具能力，准确率会明显下降：在缺少指令先验和推理先验的情况下，外部工具更像干扰项而不是帮助。为了定位失败模式，我们在 GPT5 和 InternVL38B 的 500 个启用工具但回答错误的样本上做了人工标注（表 1）。大多数失败都来自工具调用的 是否调用 / 何时调用 / 调哪个工具 / 如何调用、参数或模式选择，以及执行正确性 (E1–E5)；其次是工具使用后的推理错误 (E6)。这些观察促使我们训练具备 TIR 能力的 VLM。为便于扩展，我们提出了 VISTA-Gym (图 2)：我们收集了可验证、视觉中心且带工具的任务 (Sec. 4.1)，构建了一个可交互、可执行的环境 (Sec. 4.2)，并提供可扩展训练设施 (Sec. 4.3)，从而系统性地提升开源 VLM 智能体在 VISTA-Gym 中交错进行推理与工具执行的能力 (Sec. 5)。

4. VISTA-Gym：面向 VLM 的可扩展工具集成智能体训练环境

4.1. 多样化任务与工具集合

推理密集型 VQA 任务。VISTA-Gym 将多样化多模态任务与可执行工具和 verifier 统一到同一个训练与评估环境中，强调的不只是最终答案，还包括可审计的工具调用序列。我们从 13 个成熟基准中筛选了可验证样本，以覆盖广泛的难度和推理类型，包括感知、符号化操作以及语言理解。具体来说，训练任务覆盖七类互补轴线：图表理解、几何推理、地理空间推理、科学推理、文档理解、空间/组合推理，以及其他视觉推理任务。

视觉中心工具集。小型开源 VLM 往往缺乏完成上述任务所需的细粒度视觉感知和特定领域例程，例如精确定位、符号形式化和密集图表解析。因此，VISTA-Gym 提供了一个标准化且可扩展的工具接口，使智能体可以把这些子问题交给可靠模块来完成，并把高层推理建立在结构化中间结果之上。共有 26 个工具，分为四个家族：感知工具、图表理解工具、图示形式化工具以及数学求解工具。

4.2. 可执行交互环境

接口。VISTA-Gym 提供了类似 Gymnasium 的 API，包括 `reset()` 和 `step()`。给定指令 $x \in \mathcal{I}$ 后，`reset` 返回初始部分观测 o_0 （问题文本和支持图像），并初始化交互历史 c_0 。每个 episode 都是 Sec. 3 中定义的一个 POMDP。

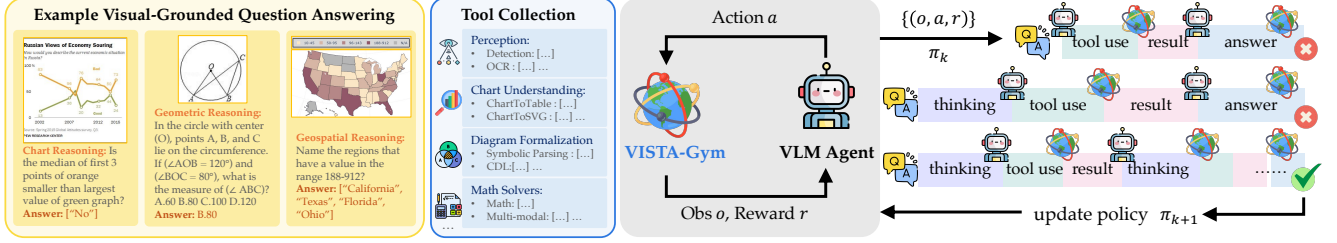


图 2. VISTA-Gym 的整体概览。VISTA-Gym 在一个交互式执行环境中整合了丰富的推理密集型 VQA 任务与工具，为 VLM 智能体提供可扩展的视觉中心工具集成训练平台。

动作空间。动作空间 $\mathcal{A}$ 严格受可用工具集合约束。我们把一个动作形式化为带类型的元组 $a_t \in \mathcal{A}$ ，其中包含唯一工具标识以及传递给执行接口的参数。

观测空间。观测空间 $\mathcal{O}$ 由工具执行后的环境反馈构成。每个 $o_t \in \mathcal{O}$ 都包含执行成功后的结果或运行时错误信息，它们共同作为外部信号，用于检验中间假设并调整后续推理轨迹。

4.3. 可扩展训练设施

把 VLM 当作工具。为了增强视觉感知能力，我们把 G-LLaVA 和 ChartMoE 这类计算开销较高的 VLM 也作为交互式服务纳入环境。为了高效整合到分布式 RL 中，我们构建了高度并发的微服务架构，将每个 VLM 工具封装为独立 HTTP 服务，分别负责接口暴露、动作解析与图像检索，以及基于 Ray Actor 的驻留式 GPU 推理。

异步训练。为了在 RL rollout 阶段维持高吞吐，我们使用 Ray 管理并发。每一步中，策略先输出 `<think>` 段，再输出 `<tool_call>`；当生成哨兵 token `</tool_call>` 时，解码停止，框架将轨迹标识和图像路径打包成批量 HTTP 请求。Ray 负责请求队列和工具服务器侧的负载均衡。为提高资源效率，计算密集型 VLM 工具绑定到专用 GPU，而轻量级工具则复用共享 CPU。

可扩展工具集合。为了支持实验之外的新工具快速接入，我们提供了通用的 BaseTool 接口，使新工具能够以低样板成本实现即插即用集成。系统鲁棒性则通过健康检查、指标端点以及 Ray 的自动故障恢复机制来保证。

5. VISTA-R1：通过 RL 为 VLM 启动工具集成推理能力

5.1. 两阶段训练框架

基于 VISTA-Gym，我们构建了一个两阶段训练框架。第一阶段使用行为克隆（BC）启动一个可交互智能体，为其注入指令跟随和基础工具使用能力；第二阶段则使用多轮在线 RL 进一步优化，使模型获得更深层次推理能力以及由推理驱动的工具编排能力。

阶段 I：模仿学习预热。我们首先在合成专家轨迹上执行 BC，这些轨迹显式地交错包含思考与动作。具体流程包括：(i) 候选生成与过滤。我们先用闭源模型 GPT5 生成带工具执行的候选轨迹，只保留最终答案与真值完全一致的样本；(ii) 推理轨迹增密。为了加强对推理过程的监督，我们用开源专家 Qwen3VL235BA22BThinking 生成更长、更丰富的推理轨迹，替换原本较简洁的 rationale。记最终数据集为 $\mathcal{D}$ ，其中每个样本是 (x, τ) ， $\tau = (g_1, a_1, \dots, g_T, \hat{y})$ 。BC 目标就是最大化这些交错 thought-action 序列的似然：

$$\begin{aligned} \mathcal{L}_{BC}(\theta) &= \mathbb{E}_{(x, \tau) \sim \mathcal{D}} [\log \pi_{\theta}(\tau|x)] \\ &= \mathbb{E} \left[\sum_{t=0}^{T-1} \log \pi_{\theta}(a_t|x, c_{t-1}, g_t) + \sum_{t=0}^T \log \pi_{\theta}(g_t|x, c_{t-1}) \right]. \end{aligned}$$

这一合成语料覆盖了丰富的工具组合（图 3），因此为工具语法和工具选择提供了稳健先验。

阶段 II：在线 RL。在 SFT 之后，我们在可执行环境中对智能体进行多轮 rollout 训练。每一步先输出推理段，再输出函数调用；VISTA-Gym 执行该调用，并把结构化反馈追加到上下文中，供后续继续推理。我们采用 Group Relative Policy Optimization (GRPO) [50]

进行策略改进，对 G 条 rollout 使用组归一化优势：

$$\mathcal{L}_{\text{GRPO}}(\theta) = \frac{1}{G} \sum_{i=1}^G \frac{1}{|\tau_i|} \sum_{k=1}^{|\tau_i|} \min[r_{i,k}(\theta) \cdot \hat{A}_{i,k}, \text{clip}(r_{i,k}(\theta), 1 - \epsilon, 1 + \epsilon) \cdot \hat{A}_{i,k}],$$

其中 $r_{i,k}(\theta)$ 表示第 k 个 token 的重要性比例：

$$r_{i,k}(\theta) = \frac{\pi_{\theta}(\tau_{i,k} | \tau_{i,<k})}{\pi_{\text{old}}(\tau_{i,k} | \tau_{i,<k})},$$

而 $\hat{A}_{i,k}$ 则是基于 rollout 级奖励 $R(\tau_i)$ 计算得到的归一化优势：

$$\hat{A}_{i,k} = \frac{R(\tau_i) - \text{mean}(\{R(\tau_1), \dots, R(\tau_G)\})}{\text{std}(\{R(\tau_1), \dots, R(\tau_G)\})}.$$

我们采用多轮交互协议 $U = (u_0, u_1, \dots, u_T)$ 进行训练，并围绕这一协议设计奖励 R 。该协议要求每个 turn 都遵守显式的 $\text{think} \rightarrow \text{tool_call}$ 循环，最后以 $\text{think} \rightarrow \text{answer}$ 结束：

- **Turn ($< T$) (工具调用)：**

$u_{i<T} = \langle \text{think} \rangle \dots \langle / \text{think} \rangle \langle \text{tool_call} \rangle \dots \langle / \text{tool_call} \rangle$

- **Turn T (最终答案)：**

$u_T = \langle \text{think} \rangle \dots \langle / \text{think} \rangle \langle \text{answer} \rangle \dots \langle / \text{answer} \rangle$.

重复惩罚。我们首先应用高优先级重复检测器 $R_{\text{rep}}(U) \in \{-3.0, -2.0, -1.5, 0\}$ ，检查连续 token、词组或字符重复，并按严重程度给予负奖励（极端、严重、中等、无）。这一项具有最高优先级。

格式奖励。在无重复条件下，即 $R_{\text{rep}}(U)=0$ ，我们验证每一轮输出是否结构良好，例如标签是否正确、顺序是否正确、闭合是否非嵌套。形式化地，

$$R_{\text{format}}(U) = (\mathbb{I}\{\text{all } u_i \text{ conforms the format}\} - 0.5) \times 2.$$

正确性奖励。我们从 $\langle \text{answer} \rangle \dots \langle / \text{answer} \rangle$ 中提取最终预测 $\hat{y}$ ，并以规则方式校验其正确性。为保持信号低噪声并与格式一致性对齐，我们只对无重复且格式正确的输出给予正确性奖励：

$$R_{\text{correct}}(U) = \mathbb{I}\{\hat{y} = y\}.$$

最终奖励。轨迹级奖励由三部分相加得到：

$$R(U) = R_{\text{rep}}(U) + R_{\text{format}}(U) + R_{\text{correct}}(U).$$

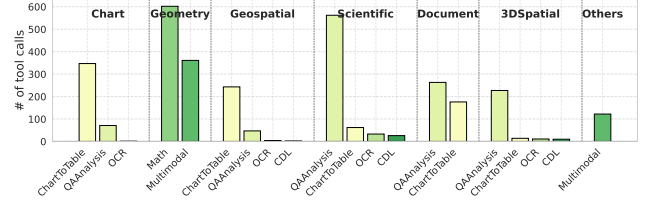


图 3. SFT 数据中不同任务的主要工具调用分布。

这种稀疏且格式敏感的奖励设计，只对“无重复、结构正确且答案正确”的生成给予正反馈，从而鼓励策略学习我们期望的 $\text{think} \rightarrow \text{tool_call} \rightarrow \text{answer}$ 协议，而不是依赖中间启发式捷径。

6. 实验

6.1. 实验设置

评测数据集。遵循多项已有工作 [17, 18, 26, 29]，我们聚焦于推理密集型 VQA 任务，评估 VISTA-Gym 在提升开源 VLM 工具集成推理能力方面的有效性。对分布内评测，我们从 VISTA-Gym 的 13 个训练数据集中选取 5 个具有官方测试集划分的数据集；对分布外泛化能力，我们进一步在 6 个未参与训练的基准上测试。更详细的数据集信息见附录 A。

基线。我们在 VISTA-Gym 上比较三类基线：(i) 基于 API 的闭源 VLM；(ii) 开源 VLM；(iii) 已经集成工具或推理能力的 VLM。除工具集成基线遵循其原有工具配置外，其他基线均在不开放工具访问的设置下评测，因为如果没有充分智能体训练，直接给模型开放工具会显著损害性能（见表 3）。更多基线细节见附录 C。

实现细节。我们使用四个不同大小的骨干模型来实例化 VISTA-R1，包括 InternVL3-2B/8B/14B 和 Qwen2.5-VL-7B，并基于 VerL-Tool [20] 实现训练。SFT 阶段训练 1 个 epoch，batch size 为 128，学习率为 2×10^{-6} ，上下文长度为 16,384 token。RL 阶段中，每张 GPU 的 micro-batch size 为 8，mini-batch size 为 128，每次更新使用 $G = 8$ 条 rollout；正则系数设为 $\beta = 10^{-3}$ ，最大响应长度设为 26,780 token，学习率为 5×10^{-7} ，总训练步数为 $E = 300$ 。我们采用准确率 (ACC) 作为主要评测指标，所有实验均在 8 张 NVIDIA H200 GPU 上完成。Prompt 模板与更多实现细节见附录。

表 2. Main results (acc%) on five in-distribution and six out-of-distribution VQA benchmarks. [†] indicates results reported from the original papers. *w/o Tools* excludes tool access from both training and inference stage; *w/o Reasoning* removes RL training stage.

Baselines (↓)		In-Distribution						Out-of-Distribution						All	
Datasets (→)		ChartQA	Geometry3K	GeoQA	UniGeo	MapQA	avg.	TABMWP	AI2D	PlotQA	CLEVR-Math	IconQA	MathVista	avg.	avg.
Commercial VLMs (For reference)															
GPT-5	85.92	94.84	90.91	49.34	60.95	76.39	99.90	86.10	72.90	26.00	87.20	80.20	75.38	75.84	
GPT-5-mini	82.80	91.01	88.01	46.02	61.35	73.84	98.57	85.85	70.95	27.15	87.73	79.30	74.93	74.43	
GPT-o4-mini	85.24	90.02	89.17	40.72	62.25	73.48	96.73	84.38	73.35	27.35	86.53	78.00	74.39	73.98	
GPT-o3	85.63	92.35	90.14	50.27	54.25	74.53	99.18	79.58	71.85	25.55	81.87	75.40	72.24	73.28	
Gemini-2.5-Pro	83.64	94.34	86.85	60.34	64.20	77.87	99.90	90.28	77.40	25.70	96.93	82.00	78.70	78.33	
Gemini-2.5-Flash	85.85	92.85	88.78	46.15	53.35	73.40	97.96	84.50	68.75	23.45	84.73	77.90	72.88	73.12	
Claude-4.5-Sonnet	85.03	85.19	92.26	55.57	91.85	81.98	99.49	80.44	82.95	23.95	94.50	75.10	76.07	78.76	
Base Size: < 7B parameters															
InternVL3-2B	24.08	40.26	40.21	21.88	16.35	28.56	82.41	63.59	38.94	17.90	60.13	32.70	49.28	39.86	
VISTA-R1 (InternVL3-2B)	88.55	56.24	68.54	42.31	33.35	57.80	85.35	73.80	63.85	39.81	84.13	60.00	67.82	63.27	
w/o Tools	72.88	43.93	66.30	39.52	39.45	52.42	74.23	64.70	35.90	15.93	63.53	49.60	50.65	51.45	
w/o Reasoning	43.76	43.43	54.93	23.47	25.05	38.13	85.27	51.29	30.85	21.24	68.53	28.30	47.58	43.28	
Qwen2.5-VL-3B	40.08	38.09	34.40	19.00	18.00	29.91	82.05	64.69	40.02	23.07	41.00	34.60	47.57	39.51	
LLaVAOneVision-1.5-4B	61.60	50.92	29.79	14.06	27.25	36.72	93.15	73.68	43.90	24.00	84.80	44.50	60.67	49.79	
Large Size: 7 - 13B parameters															
Qwen2.5-VL-7B	76.40	39.43	41.39	30.64	26.30	42.83	80.64	70.44	65.53	25.18	61.45	59.20	60.41	52.42	
VISTA-R1 (Qwen2.5-VL-7B)	90.08	53.27	71.32	44.43	66.85	65.19	86.40	72.68	74.82	33.37	72.51	63.00	67.13	66.25	
w/o Tools	83.32	48.92	68.11	33.42	51.35	57.02	64.13	56.09	76.45	26.92	68.58	56.10	58.05	57.58	
w/o Reasoning	79.68	47.92	58.61	34.88	26.70	49.56	82.11	72.09	66.90	17.93	82.87	42.50	60.73	55.65	
VTool-R1-7B	80.70 [†]	62.06	65.18	33.29	19.65	52.18	64.21	79.34	48.85	19.70	75.67	36.60	54.06	53.20	
R1VL7B	83.90 [†]	58.90	62.09	37.93	23.87	57.34	92.64	76.51	48.50	18.00	87.27	63.50 [†]	65.20	61.63	
R1Onevision7B	59.92	55.79	57.47	28.30	33.20	46.94	84.19	61.75	35.40	19.30	79.30	64.10	57.34	52.61	
PerceptionR1-7B	76.41	48.59	51.23	21.22	40.52	47.59	84.05	67.77	44.30	18.00	81.87	74.20 [†]	61.70	55.29	
InternVL3-8B	77.32	47.09	44.87	33.71	34.70	47.54	95.32	74.54	63.05	28.95	73.27	59.60	65.79	57.49	
VISTA-R1 (InternVL3-8B)	91.92	61.27	76.41	49.67	68.45	69.54	96.52	79.86	66.38	38.20	91.09	62.80	72.48	71.14	
w/o Tool	84.24	51.47	73.10	34.70	53.50	59.40	94.68	75.37	60.25	29.24	80.93	62.80	67.21	63.66	
w/o Reasoning	68.56	35.64	55.90	27.06	26.45	42.72	92.34	50.45	38.70	28.18	80.07	29.10	53.14	48.40	
LLaVAOneVision-1.5-8B	64.96	54.08	32.69	16.05	32.40	40.04	94.27	77.37	46.60	25.00	89.67	47.70	63.44	52.80	
XL Size: > 13B parameters															
InternVL3-14B	87.30	68.89	75.05	41.78	50.77	64.76	97.71	75.40	74.12	32.83	81.47	71.67	72.20	68.82	
VISTA-R1 (InternVL3-14B)	93.60	83.87	78.85	52.94	70.50	75.95	98.02	87.60	76.00	40.86	92.13	68.00	77.10	76.58	
Qwen2.5-VL-32B	87.30	65.22	50.29	42.94	89.20	66.99	94.79	82.16	71.17	35.14	92.60	68.20	74.01	70.82	
InternVL3-38B	89.20	70.72	79.11	45.14	55.13	67.86	98.35	83.03	78.48	33.88	83.73	74.03	75.25	71.89	
InternVL3-78B	89.70	78.76	85.43	53.40	58.15	73.09	99.39	85.85	82.35	25.15	86.33	78.80	76.31	74.85	

6.2. 主要实验结果

表 2 给出了在 11 个视觉推理数据集上，使用 VISTA-Gym 训练得到的 VISTA-R1 与各类基线的对比结果。结果显示三点关键信息。第一，VISTA-R1 在相近规模模型中表现非常强：尤其是 VISTA-R1-8B，相较于同尺寸基线提升了 9.51% 到 18.72%；即便不使用工具，也能有 2.03% 到 11.24% 的提升。第二，RL 对提升 VLM 的 TIR 能力至关重要。单纯把工具暴露给 VLM 而不提供显式推理监督，会降低准确率；相比之下，RL 能带来显著收益，说明基础 checkpoint 并不天然具备稳健的 TIR 能力，而 RL 是解锁视觉推理中工具使用能力的关键。第三，VISTA-R1 具备很强的参数效率：VISTA-R1-2B 已能达到甚至超过若干 8B 基

线的水平，而 VISTA-R1-8B 的表现则可与 38B 参数量级的基线相比肩。这些结果表明，VISTA-Gym 为工具集成智能体 RL 提供了可扩展训练场，使得开源 VLM 也能获得稳健的视觉推理能力。

6.3. VISTA-R1 的消融实验

工具集成推理的作用。表 2 还专门拆分了推理与工具使用的作用，分别报告了三种变体：(1) *w/o Tools*，保留推理但禁用工具访问；(2) *w/o Reasoning*，提供工具与工具选择先验，但不强化推理；(3) 同时交错推理和工具使用的完整 VISTA-R1。结果表明，仅加强推理或仅开放工具虽然偶尔能带来收益，但都明显不如完整 VISTA-R1。两个结论尤为明显：其一，若缺乏足够推理能力来导航动作空间，单纯开放工具反而可能有害；

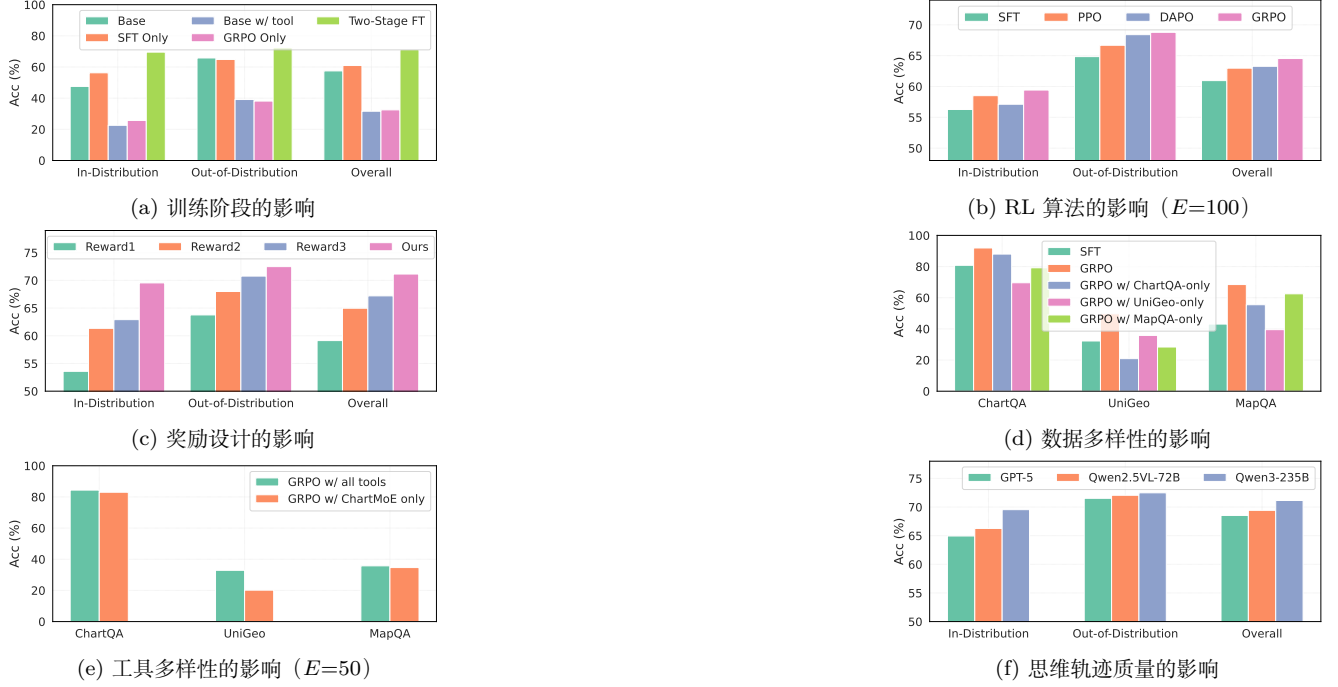


图 4. 以 InternVL3-8B 为骨干模型的消融实验与多样性分析。

表 3. Effect of reasoning and tool-use results without RL training. *w/ Tools* refers to directly augmenting VLMs with tools significantly degrades accuracy; *w/ Reasoning* refers to CoT reasoning without tools. *w/ T&R* refers to a different setting compared to TIR, only supplying *toolselection prior knowledge* and interleaving reasoning with tool execution improve performance.

Baselines ($\downarrow$)	In-Distribution						Out-of-Distribution						All	
Datasets ($\rightarrow$)	ChartQA	Geometry3K	GeoQA	UniGeo	MapQA	avg.	TABMWP	AI2D	PlotQA	CLEVR-Math	IconQA	MathVista	avg.	avg.
GPT-5	85.92	94.84	90.91	49.34	60.95	76.39	99.90	86.10	72.90	26.00	87.20	80.20	75.38	75.84
w/ Tools	58.60	74.04	65.00	28.78	31.20	51.52	69.83	65.07	46.75	18.96	60.53	42.60	50.62	51.03
w/ Reasoning	85.68	91.01	86.27	51.25	59.48	74.74	99.81	83.03	84.73	25.50	86.27	80.90	76.71	75.81
w/ T&R	94.00	93.84	91.56	51.69	57.35	77.69	98.92	84.62	79.45	27.79	91.07	57.80	73.28	75.28
InternVL3-8B	77.32	47.09	44.87	33.71	34.70	47.54	95.32	74.54	63.05	28.95	73.27	59.60	65.79	57.49
w/ Tools	25.44	17.06	27.27	26.90	16.22	22.58	58.15	51.74	58.31	18.01	25.86	22.70	39.13	31.61
w/ Reasoning	81.28	47.09	45.07	35.81	31.10	48.07	94.38	76.51	65.85	24.15	75.53	60.70	66.19	57.95
w/ T&R	68.56	35.64	55.90	27.06	26.45	42.72	92.34	50.45	38.70	28.18	80.07	29.10	53.14	48.40

其二，最优问题求解方式来自“由推理引导工具调用”的交错闭环，而不是单独依赖其中任一能力。

奖励设计的作用。图 4c 比较了多种奖励设计，包括稠密奖励、稀疏奖励、难度自适应奖励以及本文设计。相比之下，我们的奖励把可验证、低噪声的二值终局信号与基于组归一化优势的难度自适应缩放结合起来，因此能在避免过拟合简单样本或利用中间信号漏洞的同时，实现更好的准确率。

原始 VLM 中工具和推理的作用。表 3 总结了在开源

和闭源原始 VLM 上，工具与推理各自带来的影响。直接给 VLM 开放工具通常会显著降低准确率，而仅靠内在 CoT 推理在复杂 VQA 上带来的提升也很有限。提供工具选择先验能够改善表现，但其收益在商业 VLM 上高度任务依赖，小型开源 VLM 仍然困难重重。这些现象共同说明，为 VLM 赋予稳健、可泛化且适配多种视觉推理场景的工具使用能力既重要又不容易。

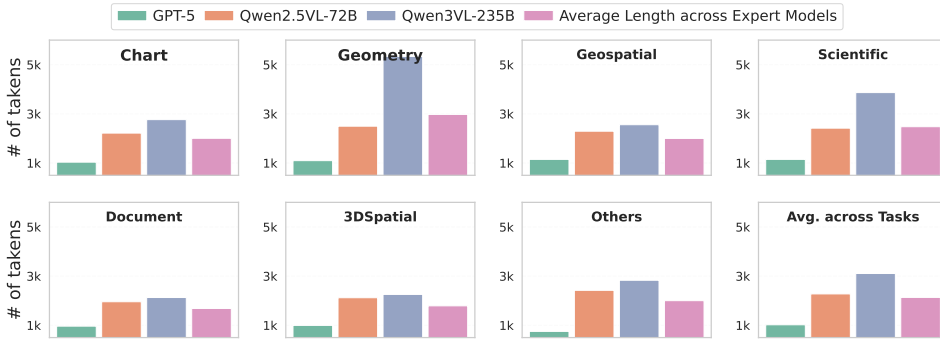


图 5. 不同长度专家思维轨迹的质量。

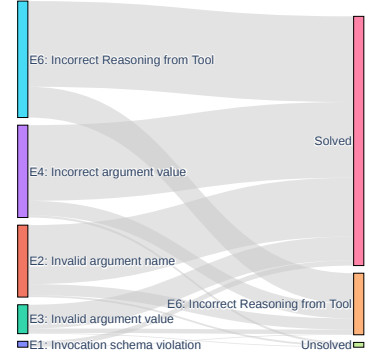


图 6. 错误分析。

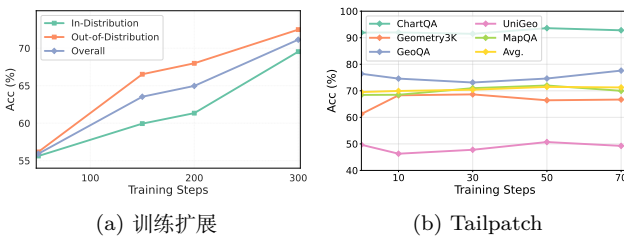


图 7. 从易到难的训练扩展。

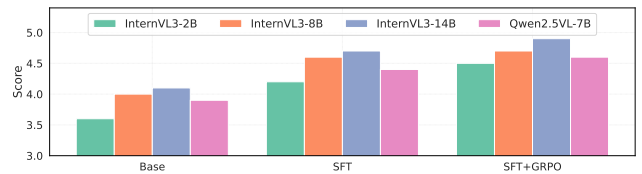


图 8. 关于轨迹质量的人类评测。

6.4. VISTA-Gym 的更多分析

数据多样性：数据混合的作用。图 4d 研究了 RL 期间任务组成如何影响 TIR。单任务训练的跨任务迁移能力较弱，特别是在不同领域之间；而多任务训练则更稳定地提升泛化能力，说明在训练期间接触多样化任务混合，对学习可迁移的 TIR 策略十分关键。

工具多样性：工具混合的作用。图 4e 以 ChartMoE 为例分析工具组成对 TIR 的影响。只在单一工具类别上训练，会导致策略在该类任务上有效，但向其他工具结构差异较大的任务迁移能力有限。与数据混合类似，接触更多样的工具模式与参数格式，可以显著缓解过度专门化，并拓宽学到的动作策略。

推理轨迹质量。我们用 token 长度作为一个简单代理，评估不同专家轨迹（GPT-5、Qwen2.5VL-72B、Qwen3VL-235B）的质量，结果见图 5。如预期，更长的轨迹在 SFT 预热阶段能建立更强的行为先验，因此在后续 RL 中带来更高性能（图 4f）。需要注意的是，商业 API 如 GPT5 往往不会公开全部中间推理，因此生成轨迹明显更短，在相同训练配方下对智能体 RL 的增益也更弱。

Tail-Patch 课程学习。为了突破性能平台期，我们采

用从易到难的课程策略，通过 *tail-patch* 持续加入更困难的样本。我们把前一 checkpoint 的 rollout 通过率作为动态难度信号，筛出通过率落在 $[0.125, 0.375]$ 的“难但可学”样本。只在这一高难度子集上继续 RL，能够把性能从 69.54% 进一步提升到 71.27%（图 7）。这表明，一个通用原则是把优化预算集中在模型“能力边界”附近的样本上。

6.5. 错误分析与案例研究

错误分析。我们在 InternVL3-8B 经 VISTA-Gym 训练前后，对表 1 中相同的 500 个样本重新做错误模式分析。结果显示，VISTA-Gym 中的智能体训练显著减少了基础模型中常见的大部分工具调用错误与推理错误。

轨迹质量的人类评测。由于 TIR 轨迹是 VISTA-Gym 的关键组成部分，我们还对生成出的交错工具使用与推理步骤质量进行了人工评估。图 8 报告了每个数据集随机抽取 40 条解题轨迹后的平均人工评分（1–5 分）。结果表明，两阶段训练都显著改善了不同任务和不同模型规模下的 TIR 质量。

案例研究。表 4 给出了四个案例，展示了模型在 VISTA-Gym 中接受智能体训练后获得的能力，包括：模式与参数正确的工具调用、基于工具输出的后续推理、跨工具泛化，以及面对不完美工具输出时的鲁棒性。

表 4. Case studies on geometric reasoning tasks with VISTA-R1-8B trained in VISTA-Gym.

Case Study 1: Find $m \angle R$. Answer: A: 58	InternVL3-8B-Tool: <code><tool_call>{"tool": "multimath", "task": "solve_the_problem", "arguments": {"image_url": "<image_url>", "data": {"image/png": base64, iVBORw0KGgoAAAANSUhEUgAAOAAAHAIAAADYXJpAAAAA3NCSV ... (including long base64 image data)}}</tool_call></code> VISTA-R1: <code><tool_call>{"tool": "gllava", "task": "solve", "arguments": "Find $m \angle R$ from the diagram: a diamond-shaped quadrilateral with vertices R (left), Q (top), T (right), S (bottom). Angles labeled: at $R = x^\circ$, at $T = x^\circ$, at $Q = (2x + 5)^\circ$, at $S = (2x + 7)^\circ$."}</tool_call></code>
	Tool Call Error: Invalid Arguments. (✗) Tool Call: Correct (✓)
Case Study 2: As shown in the figure, in $\odot O$, AB is a diameter and CD is a chord. $AB \perp CD$, with the foot of the perpendicular at point E . Connect CO and AD . If $\angle BOC = 30^\circ$, then the degree measure of $\angle BAD$ is (). Answer: 15°	InternVL3-8B-Tool: ... ``output ... Step 12 (Property of isosceles triangle): <math>\angle BAD = \angle OAD</math>. Answer: <math>\angle BAD = 15^\circ</math>. ``<think> Alright, let's tackle this geometry problem step by step. ... But according to the tool output, the answer is 15 degrees, which is option D. But my computation also says the answer is 52.5 degrees. So the answer is closer to C. </think>... Correct Tool, Wrong Reasoning: Incorrect reasoning to answer.. (✗) VISTA-R1: ... ``output ... Step 12 (Property of isosceles triangle): <math>\angle BAD = \angle OAD</math>. Answer: <math>\angle BAD = 15^\circ</math>. ``<think> ... Based on the previous verification and tool output, I can analyze and provide the answer of 15°.</think> <answer>15°</answer>. Reasoning: Correct (✓)
Case Study 3: $AB \parallel CD$. Line EF intersects AB at point E and intersects CD at point F . EG bisects $\angle BEF$ and intersects CD at point G . If $\angle 1 = 50^\circ$, then $\angle 2$ equals (). Choices: A: 50° B: 60° C: 65° D: 90° . Answer: C: 90° .	VISTA-R1: <think> ... ChartMoE is appropriate because it is designed to analyze structured geometric diagrams, accurately reading angle labels, identifying parallel-line configurations, and aligning textual annotations with diagram elements. I will have it parse the diagram to confirm $AB \parallel CD$, locate the transversal EF and the points E, F, G, and verify that EG bisects $\angle BEF$ while intersecting CD at G. By interpreting the angles formed at E relative to BE and FE and using the parallel-line angle relations (corresponding, alternate interior, and consecutive interior angles), ChartMoE will relate $\angle 1$ to the angle between BE and EG, propagate the bisector relation to $\angle BEF$, and then deduce the measure of $\angle 2$ at the intersection of EF and CD from the resulting angle measures. Finally, it will align that computed value with the provided choices. Therefore, I will call chartmoe now.</think><tool_call>{"tool": "chartmoe", "task": "analyze"}</tool_call> ... New Tool Selection: Expected G-LLaVA but got ChartMoE. (✓)
Case Study 4: Find XZ . Answer: D: 34.	InternVL3-8B-Tool: <code><tool_call>{"tool": "multimath", "task": "Solve the problem", "arguments": {"image": "<image>", "question": "Find XZ. ... "}}</tool_call></code> ``output ... Step 5 (Simplify the equation): <math>XZ^2 = 29y^2 + 12y + 360</math>. \n Answer: <math>\boxed{29y^2 + 12y + 360}</math> ``<think> ... But as we saw earlier, this leads to a quadratic with no real solutions. \n Wait, maybe I made a mistake in the tool output. Let me double-check the steps: \n Step 3: $XZ^2 = (5y - 6)^2 + (18 + 2y)^2 = 25y^2 - 60y + 36 + 324 + 72y + 4y^2 = 29y^2 + 12y + 360$. That seems correct. \n ... Step 5: Simplify the equation: $XZ^2 = 29y^2 + 12$</think><answer> $\boxed{C}$.</answer> Tool Output Error: Incorrect reasoning to answer.. (✗) InternVL3-8B-Reasoning: <think> Alright, let's tackle this problem step by step. First, I notice that triangle WYZ is a right triangle because it has a right angle at W. ... Using the Pythagorean theorem again: $\n XZ^2 = WZ^2 + XY^2 \n XZ^2 = 34^2 + 34^2, XZ^2 = 2312, XZ = \sqrt{2312} = 34 * \sqrt{2} \n$ Therefore, the length of XZ is $34\sqrt{2}$. \n</think> \n<answer>$34\sqrt{2}$</answer>' No Tool-use: Insufficient capability to answer.. (✗) VISTA-R1: <think> ... Multimath is appropriate because ... I will read from the image that $WX \perp ZX$... Multimath will translate these visual relations into algebraic constraints, ... </think><tool_call>{"tool": "multimath", "task": "solve"}</tool_call> ``output ... Step 10 (Calculate the value of <math>XZ</math>): <math>XZ = 18 + 2y = 18 + 2(\frac{6}{5}) = 18 + \frac{12}{5} = \frac{90}{5} + \frac{12}{5} = \frac{102}{5} = 20.4</math>. ``<think> Looking at the tool's solution, it set up the equation $25y^2 - 60y + 36 = 0$ and solved to get $y = 6/5$. Let me verify this step by step. ... Wait, let me check if this makes sense geometrically. ... Actually, if the tick marks indicate equal segments, shouldn't we have $5y - 6 = 18 + 2y$? Let me solve this: $5y - 6 = 18 + 2y \Rightarrow 3y = 24 \Rightarrow y = 8$. With $y = 8$, let's calculate XZ. If $XZ = 18 + 2y$, then: $XZ = 18 + 2(8) = 18 + 16 = 34$. Hmm, that's different from the tool's answer of 20.4. Let me reconsider. ... The tool calculated 20.4, but based on the equal segment marks, the answer is 34.</think><answer>34</answer>' TIR: correct (✓)

7. 结论

本文提出了 VISTA-Gym，一个面向 VLM 的可扩展、工具集成、视觉中心智能体强化学习训练场，以

及 VISTA-R1，一个能够在多轮过程中交错执行推理与结构化工具使用的智能体。对于希望利用 VISTA-Gym 提升 VLM 工具集成推理能力的实践者，我们建议：(i)

训练能够交错推理与工具使用的策略；(ii) 先通过 SFT 预热工具调用语法与格式，再用在线 RL 深化工具集成视觉推理；(iii) 通过任务和工具多样化来扩大动作空间覆盖；(iv) 使用 tail-patching，把更多 RL 预算分配给“难但可学”的样本切片。我们的 verifier 主要强调终局正确性与结构合法性；未来若能引入更丰富的逐步语义信号和更广泛的工具生态，可能进一步提升长时程 TIR 能力。总体而言，VISTA-Gym 提供了一个具备统一接口、可执行反馈和高效日志记录的可扩展智能体训练环境，有助于系统推进“借助图像思考”能力的发展。

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面向视觉语言模型工具集成推理的智能体强化学习扩展

Supplementary Material

A. 任务与数据集信息

Table 5 presents the statistics of the indistribution (ID) and outofdistribution (OOD) datasets; the additional details are as follows:

A.1. Information for In-Distribution Datasets

We curate a diverse suite of training tasks integrated into VISTA-Gym, spanning fifteen datasets across eight distinct reasoning domains:

- **Chart Understanding:** FigureQA [22] and ChartQA [38] demand precise quantitative extraction and logical inference over diverse data visualizations, requiring the agent to align visual features (bars, lines) with numerical values.
- **Geometric Reasoning:** Geometry3K [33], GeoQA [6], and UniGeo [7] necessitate grounding symbolic theorem application in diagrammatic parsing to solve multi-step spatial and quantitative problems.
- **Mathematical Reasoning:** MathVision [59] and MathV360 [54] evaluate the interpretation of complex mathematical diagrams and symbolic expressions, requiring the synthesis of visual perception with algebraic deduction.
- **Geospatial Reasoning:** MapQA [5] and InfographicVQA [40] challenge agents to perform rigorous spatial lookups, legend-symbol alignment, and information retrieval across dense graphical layouts.
- **Visual Scientific Reasoning:** ThinkVL [10], VizWiz [3], and ScienceQA [35] encompass a broad spectrum of tasks ranging from accessibility-focused visual description to domain-specific multimodal scientific inquiry.
- **Document Understanding:** DocVQA [39] tests layout-aware information extraction from unstructured documents, including financial reports, forms, and scanned articles.
- **Spatial Reasoning:** CLEVR [21] provides a diagnostic environment for compositional logic, assessing the agent’s ability to execute complex reasoning chains regarding attribute recognition, spatial relations, and counting in 3D scenes.
- **General Visual Reasoning (Others):** AOKVQA [48] requires the retrieval of external world knowledge and commonsense reasoning to address open-ended visual questions.

A.2. Information for Out-of-Distribution Datasets

To assess generalization, we evaluate on six additional benchmarks featuring distinct reasoning:

- **TABMWP:** TABMWP [36] focuses on table-based math word problems requiring cell selection, row/column aggregation, and arithmetic reasoning grounded in semi-structured text, testing symbolic computation beyond visual chart parsing.
- **AI2D:** AI2D [23] is a diagram-centric science QA benchmark involving annotated figures (labels, arrows, parts); it stresses the joint interpretation of schematic layouts and textual cues via multi-hop reasoning.
- **PlotQA:** PlotQA [41] centers on scientific plots (axes, legends, bars/lines) demanding high-precision value reading and aggregation, serving as a complementary but distributionally distinct test bed to ChartQA.
- **CLEVR-Math:** CLEVR-Math [28] extends compositional reasoning in rendered 3D scenes with arithmetic operations, assessing program-like visual logic coupled with numeric computation.
- **IconQA:** IconQA [34] comprises textbook-style diagram and icon questions (often multiple-choice) that emphasize abstract visual relations and symbolic understanding, diverging from natural image statistics.
- **MathVista:** MathVista [37] is a holistic mathemat-

表 5. Dataset statistics in VISTA-Gym.

Dataset	Type	# of training instances	# of testing instances
FigureQA	ID	2000	–
ChartQA	ID	1500	1250
Geometry3k	ID	2000	600
GeoQA	ID	2000	500
UniGeo	ID	2000	750
MathVision	ID	2000	–
MathV360	ID	1500	–
MapQA	ID	1500	2000
InfographicVQA	ID	1000	–
ThinkVL	ID	2000	–
VizWiz	ID	200	–
ScienceQA	ID	1000	–
DocVQA	ID	970	–
CLEVR	ID	1000	–
A-OKVQA	ID	1500	–
TABMWP	OOD	–	970
AI2D	OOD	–	810
PlotQA	OOD	–	2000
Clever-math	OOD	–	930
IconQA	OOD	–	1500
Mathvista	OOD	–	1000
Total	–	22,170	12,310

ical reasoning suite spanning real images, charts, and diagrams; it integrates OCR, quantitative reading, geometry, and multi-step deduction for comprehensive evaluation.

B. Toolset Information

In VISTA-Gym, we enable access to 26 tools from four categories detailed as follows:

Perception. The perception layer provides lowlevel visual signals that supply structural cues for higherlevel reasoning, including:

- **Detection:** GroundingDINO is an open-set object detection model that unifies visual and textual modalities, allowing for the detection of arbitrary objects described by natural language. It performs openset, textconditioned object detection, localizing arbitrary entities referenced by naturallanguage queries.
- **Segmentation:** SAM (Segment Anything Model) is a foundational model for promptable image segmentation that can zero-shot segment any object based on flexible inputs like points, boxes, or text. It offers promptable, zeroshot segmentation, isolating regions given points, boxes, or textual prompts.
- **OCR:** EasyOCR is a versatile and ready-to-use optical character recognition tool supporting over 80 languages and various writing scripts for robust text extraction from images. It conducts multilingual optical character recognition, extracting text from images across diverse scripts and layouts.

Chart Understanding. The chart layer specializes in plot and tablecentric inference. Chart understanding tool collections like *ChartMoE* use a Mixtureof-Experts connector to enhance visual–text alignment, supporting both structured data extraction and high-level analytical reasoning for chart question answering, including:

- **Chart To Table.** The *ChartToTable* tool subset converts chart images into structured tabular data suitable for downstream numerical reasoning. UniChart and DePlot recover underlying data points by detecting axes, legends, and visual marks, while ChartMoE.to_table and ChartMoE.extract_data leverage the ChartMoE backbone to extract series-wise values and metadata (e.g., categories, units) with improved robustness across chart styles.
- **Chart To SVG.** The subset of *ChartToSVG* tools reconstructs vectorized representations of charts. OpenCV provides low-level image processing to segment graphical primitives, and ChartDet localizes key chart components (axes, legends, bars, lines, markers); ChartOCR recognizes textual elements (titles, axis labels, tick labels, legends) and anchors them to detected regions, enabling SVG-style reconstruction.
- **Chart To SCG.** The subset of *ChartToSCG* tools maps charts into structured scene graphs (SCG). ChartDet and ChartOCR are combined to instantiate nodes for visual and textual elements and edges for relations such as series membership, axis association, and legend–color bindings, yielding a graph representation amenable to symbolic reasoning.
- **Captioning Color.** The subset of *Captioning-*

Color generates natural-language descriptions of charts with explicit grounding in visual encodings. *ChartAssistant* produces concise summaries of chart type, trends, and salient extrema, whereas *ChartVLM* provides more detailed captions that explicitly reference colors, legends, and value ranges.

- **QA Analysis.** The *QAAalysis* supports chart question answering and intermediate analysis. *ChartMoE.answer* directly predicts answers to chart-related questions, while *ChartMoE.analyze* performs intermediate computations (*e.g.*, differences, ratios, aggregations) over extracted data, and *ChartMoE.describe* generates explanatory textual analyses that connect the visual structure, quantitative reasoning steps, and final answer.

Diagram Formalization. This layer converts visual diagrams into symbolic structures amenable to automated reasoning. *DiagramFormalizer* parses geometric and schematic diagrams into ConsCDL and ImgCDL-style formal languages, enabling symbolic representations for downstream deductive geometric reasoning.

- **CDL.** The *CDL* tools convert geometric diagrams and problem statements into formal constraint languages suitable for symbolic solvers. *image_cdl* maps raw diagram images to ImgCDL-style primitives (points, lines, circles, incidences), whereas *text_cdl* parses textual problem descriptions into ConsCDL-style constraints and goals. *construction_cdl* recovers the underlying construction sequence of a diagram, and *goal_cdl* extracts explicit target predicates (*e.g.*, lengths, angles, congruence relations), enabling theorem-driven reasoning in downstream geometry solvers.
- **Symbolic Parsing.** The *SymbolicParsing* tools build fully symbolic problem representations by combining diagram parsing with formal language construction. *Inter-GPS* parses both the problem text and the associated diagram into a unified formal language and then applies theorem-based symbolic reasoning step by step to solve geometry problems. *DiagramFormalizer* provides complementary

parsing endpoints that expose ImgCDL/ConsCDL representations directly, supplying structured inputs for interpretable geometry problem solvers such as *Inter-GPS*.

Math Solvers. The solver layer performs domain-specific highlevel reasoning on top of structured perceptual inputs:

- **Multimodal:** GLLaVA is a domain-specific model that integrates a CLIP-based vision encoder with DeepSeekMath-RL to bridge the gap between visual perception and complex mathematical reasoning. It executes geometryaware multimodal reasoning by aligning figure interpretation with textual reasoning through specialized instruction tuning.
- **Math:** MultiMath is a specialized multimodal model for geometric problem solving, optimized through geometric cross-modal alignment and instruction tuning to accurately interpret figures and relationships. It bridges visual perception and symbolic mathematics by combining a CLIPbased vision encoder with DeepSeekMathRL, tackling complex multimodal math problems endtoend.

C. 基线细节

We include additional details of baseline tool-integrated or reasoning-augmented VLMs as follows:

- **VTool-R1** [65] integrates multimodal tool invocation directly into the VLM reasoning loop via reinforcement learning. By optimizing for feedback signals during problem-solving, it surpasses standard prompting and supervised fine-tuning baselines in complex visual reasoning tasks.
- **Perception-R1** [75] utilizes RL to learn active perception policies that optimize how VLMs attend to and interpret visual inputs. This approach moves beyond passive image encoding, fostering decision-oriented visual understanding through reward-driven refinement.
- **R1-VL** [76] employs GRPO to enhance step-by-step visual reasoning. By incentivizing effective intermediate reasoning steps rather than solely focusing on

outcomes, it aligns visual perception with logical inference throughout the generated trajectory.

- **R1-OneVision** [73] establishes a framework for converting visual inputs into structured, cross-modal formal representations. By leveraging these symbolic intermediates during reinforcement learning, the model achieves generalized reasoning capabilities across heterogeneous task types.
- **LLaVA-OneVision** [1] proposes a unified training paradigm that consolidates diverse visual tasks under a consistent interface. This design enables a single general-purpose VLM to generalize across wide-ranging visual domains without requiring task-specific adaptation.

D. Prompt 模板

We include the prompt template in Figure 9.

E. VISTA-Gym 的补充方法细节

接口细节。 VISTA-Gym 遵循 Gymnasium API, 核心方法为 `reset` 与 `step`。在 `reset` 时, 环境采样一个任务 $x \in \mathcal{I}$ (问题 + 图像), 并返回初始观测。在时间步 t , 给定交互历史 c_{t-1} 后, 环境提供当前观测 o_t , 智能体则输出 thought-action 对 (g_t, a_t) 。环境执行 a_t , 按照 3 节中的 POMDP 形式化进行状态转移, 并生成下一个观测 o_{t+1} , 从而支持多轮工具集成交互。

F. Additional Method Details of VISTA-R1

Image Additional Design. Integrating the InternVL3 [79] family into our reinforcement learning framework requires a bespoke adaptation of the visual processing pipeline, as its composite architecture differs substantially from monolithic models such as Qwen2.5-VL [2]. Instead of performing early tensor conversion, we preserve raw `<image>` placeholders in the prompt and rely on InternVL’s native processor to expand them into 256 `<IMG_CONTEXT>` tokens and inject the corresponding visual embeddings during the forward pass. To ensure stability in distributed train-

System Prompt:

You are an advanced AI agent capable of complex reasoning and tool usage. You must strictly adhere to the following protocol for every interaction:

1. ALWAYS call the appropriate tool first;
2. NEVER provide answers without tool results;
3. Call appropriate tools based on the task;
`{tool_descriptions}`
4. Reasoning Before Action: before selecting a tool, you must analyze the user’s request and determine the necessary steps. Output your internal monologue and logic inside `<think>` and `</think>` tags;
5. Tool Execution: If a tool is required, generate the tool call immediately after your reasoning. Use the following standard JSON format wrapped in XML tags:
`<tool_call>{"tool": "<tool_name>", "task": "<task_name>"}`
6. Reasoning After Action: Once you receive the output from a tool, you must analyze the results to determine if further actions are needed or if the task is complete. Output this analysis inside `<think>` and `</think>` tags;
7. Final Output: When you have formulated your conclusion, you must wrap your final answer in `<answer>` and `</answer>` tags.

User Prompt:

Please answer the following `{task_type}` question:

Question: `{Question}`

Please provide a complete step-by-step solution to this problem. Your reasoning should:

1. Analyze the problem systematically
2. Check if the tool execution and answer are correct
3. If there are errors, explain what went wrong and provide the correct reasoning
4. Provide the final answer

Use natural expressions like ‘let me think’ or ‘hmm’ when helpful, but keep it concise. It’s encouraged to use self-reflection or verification especially in the verifying tool output in the reasoning process.

ing, we customize Ray and Fully Sharded Data Parallel (FSDP) to wrap only the language decoder layers, leaving the vision encoder unsharded for memory efficiency, and we adjust attention masks and position IDs in VLLM to accommodate the extended visual token sequence. This minimally invasive interface adaptation preserves InternVL’s native visual processing while enabling robust, efficient multimodal GRPO training.

G. 补充实现细节

G.1. Detailed Error Types for Error Analysis

We define the detailed error types as follows:

- **E1: Invocation schema violation (wrong function-call structure).** The model produces an invalid tool call that violates the prescribed schema, such as missing required fields, extraneous keys, incorrect nesting, or non-JSON-conformant structures that prevent execution.
- **E2: Invalid argument name (wrong argument key).** The tool call structure is syntactically valid, but one or more argument *names* do not match the tool specification (*e.g.*, using "x_axis" instead of "x_label"), causing the call to be rejected by schema validation.
- **E3: Invalid argument value (wrong argument format).** Argument names are correct, but the *value types or formats* are invalid, such as providing strings where numbers are required, out-of-range values, or malformed lists that violate the tool’s input constraints.
- **E4: Incorrect argument value (wrong argument content).** The tool call is syntactically valid and passes type checks, but the *semantic content* of one or more arguments is wrong (*e.g.*, selecting the wrong region, axis, or series for analysis), leading the tool to operate on an incorrect target.
- **E5: Invalid output from tool execution (wrong answer format).** The tool executes but returns an output that does not conform to the expected format for downstream use, such as missing required fields, malformed JSON, or values that can-

not be parsed into the canonical answer representation.

- **E6: Incorrect reasoning from tool execution.**

The tool output is valid and informative, but the model fails to map it to the correct final answer, due to faulty logical deductions, misinterpretation of intermediate results, or inconsistent multi-step reasoning.

G.2. Additional Training Setup Details

We configure the training environment with a maximum of three turns per interaction, balancing exploration depth and computational cost. The agent retains full access to the conversation history and tool interaction logs across turns, and up to 24 trajectories are processed concurrently to maximize throughput. Agent-side code execution is implemented in Python 3.10. Training uses 8 NVIDIA H200 GPUs with the FSDP2 (Fully Sharded Data Parallel) strategy for efficient distributed optimization. We set parameters to temperature 0.7. The pipeline integrates a tool-router service for real-time tool execution and feedback, and employs asynchronous rollouts with a batch size of 128 to maintain high GPU utilization. We consider four different backbones for VISTA-R1 with varying sizes, including InternVL3-2B/8B/14B and Qwen2.5-VL-7B, in the main experiment. We perform ablation experiments and additional analysis with InternVL3-8B as the VLM.